

LECTURE 2 :

2.1 Structure of HTML Web Page

2.2 Usage of HTML Element

2.3 Special Characters & Grouping Elements





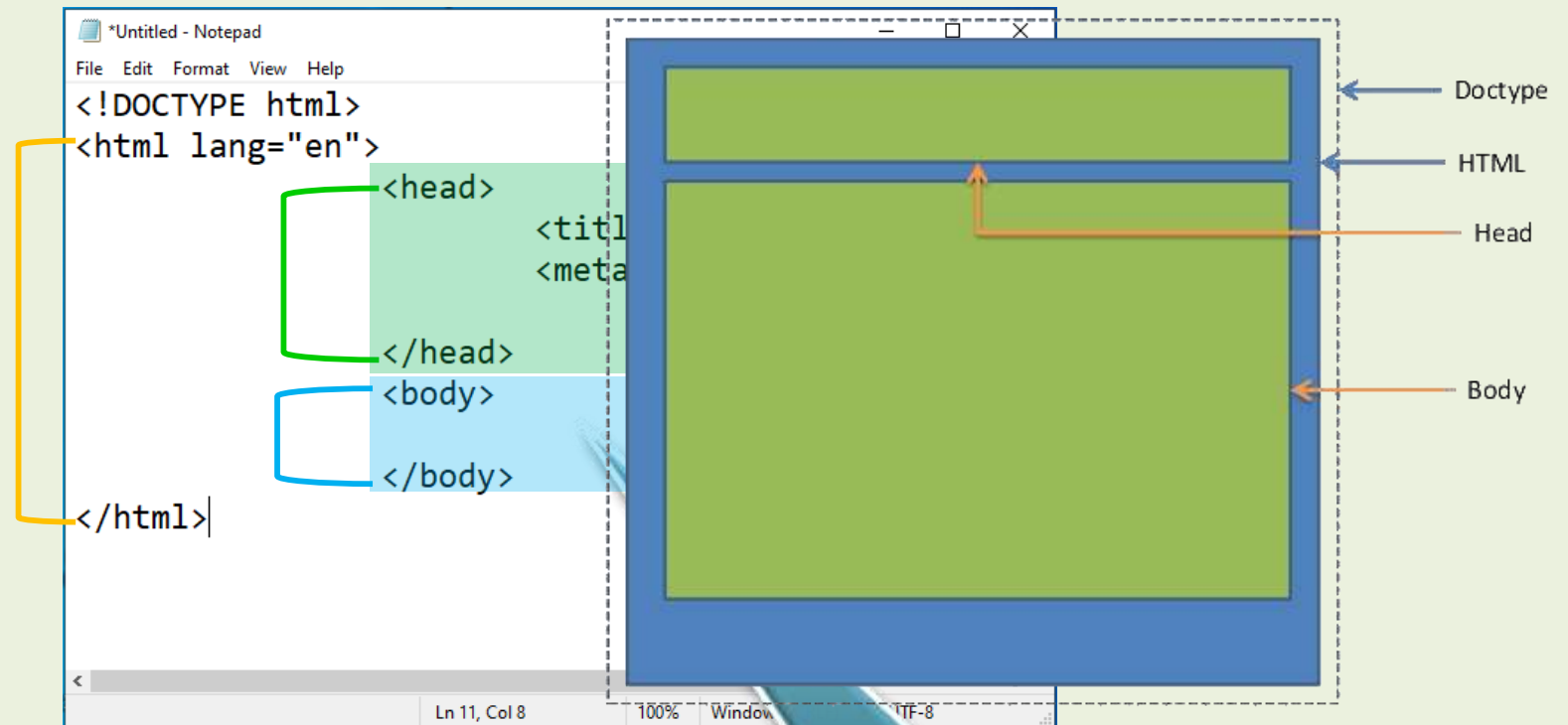
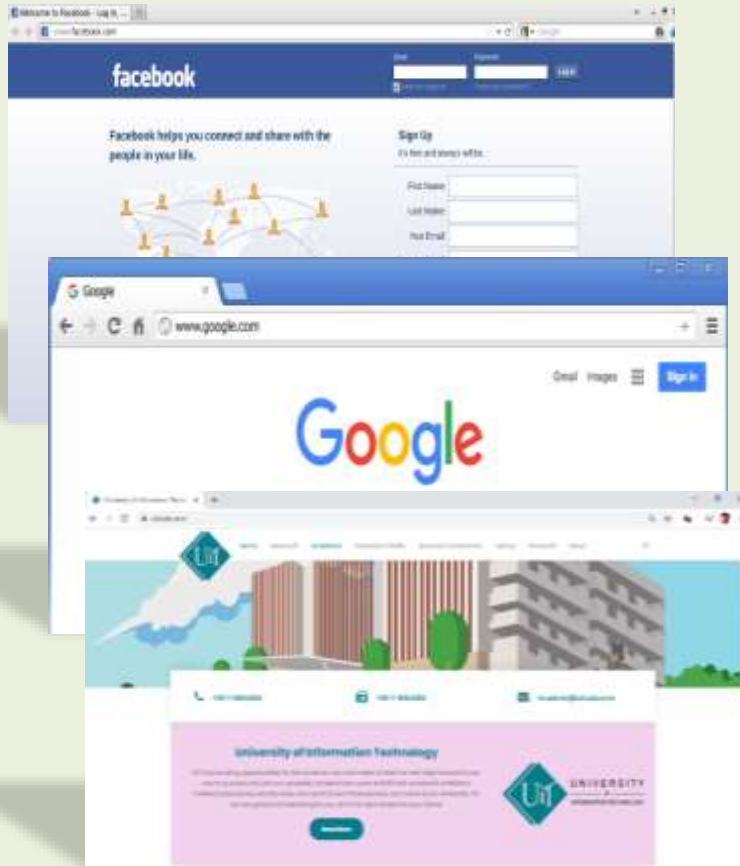
2.1

Structure of HTML Web Page

About HTML5 Web Page

- ✓ HTML can be written in any simple editor like *Notepad*, *Adobe Dreamweaver*, etc...
- ✓ **.html** or **.htm** are extensions used for saving HTML file.
- ✓ Web browsers (Chrome, Firefox, etc.) are able to read HTML and to display webpage as output.

Basic HTML5 Web Page Template

A diagram illustrating the structure of an HTML5 web page template. On the left, a Notepad window shows the code:

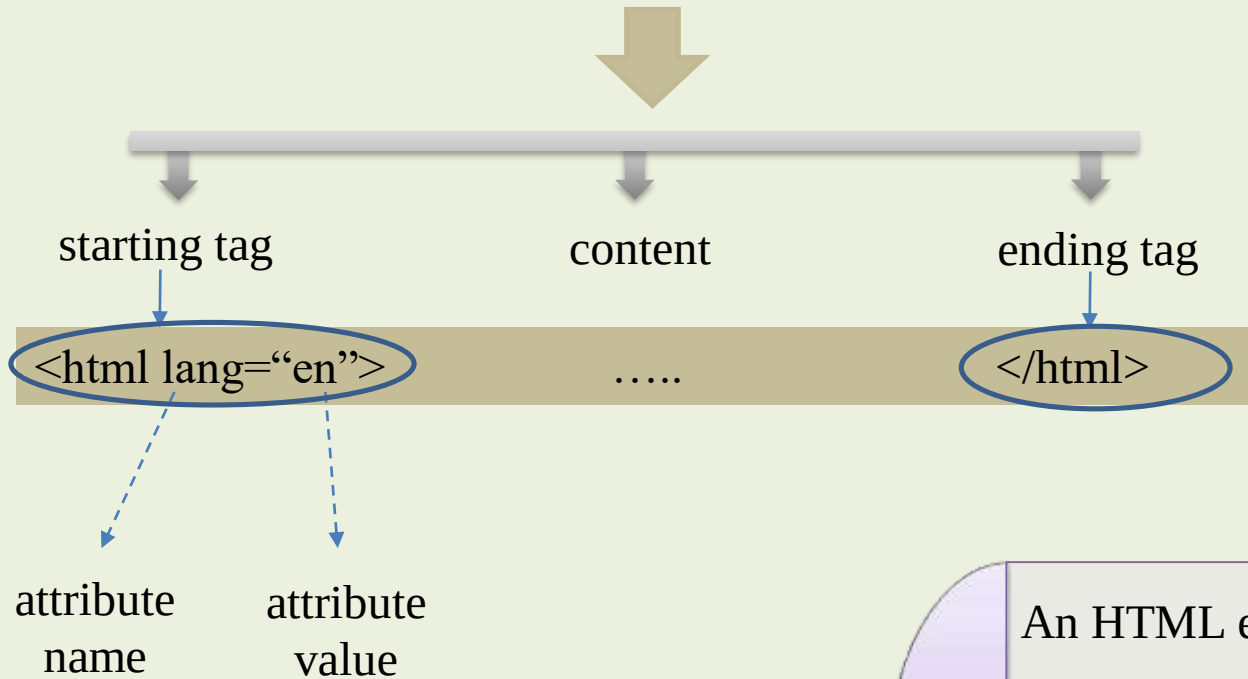
```
*Untitled - Notepad
File Edit Format View Help
<!DOCTYPE html>
<html lang="en">
  <head>
    <title>
    <meta>
  </head>
  <body>
  </body>
</html>
```

 On the right, a visual representation of the page structure is shown as a nested box. The outermost box is blue and labeled 'HTML'. Inside it is a green box labeled 'Doctype'. Inside the green box is a light green box labeled 'Head'. Inside the light green box is a light blue box labeled 'Body'. Arrows point from the labels to their respective boxes. A blue arrow points from the Notepad window to the visual representation.

Body text and more
HTML5 tags go here!!!

HTML Elements

Element



`<html> ... </html>`

`<head> ... </head>`

`<title> ... </title>`

`<meta>`

`<br>`

An HTML element is an individual component of an HTML document.

It is defined by a starting tag with an ending tag if other content contains.

Some HTML elements don't need to be closed (called void elements).

Basic HTML5 Web Page Template

There are two sections on a web page:

Head Section

1

contains **information**
that describes the web
page document

Body Section

2

contains actual tags, text,
images and other objects
that are displayed by the
browser as a web page

The HTML **<head>** Element

<head>

It is a
container
for all
the head
elements
and can
include

a **title** for
the
document

scripts

styles

meta
information

and more...

The HTML **<title>** Element

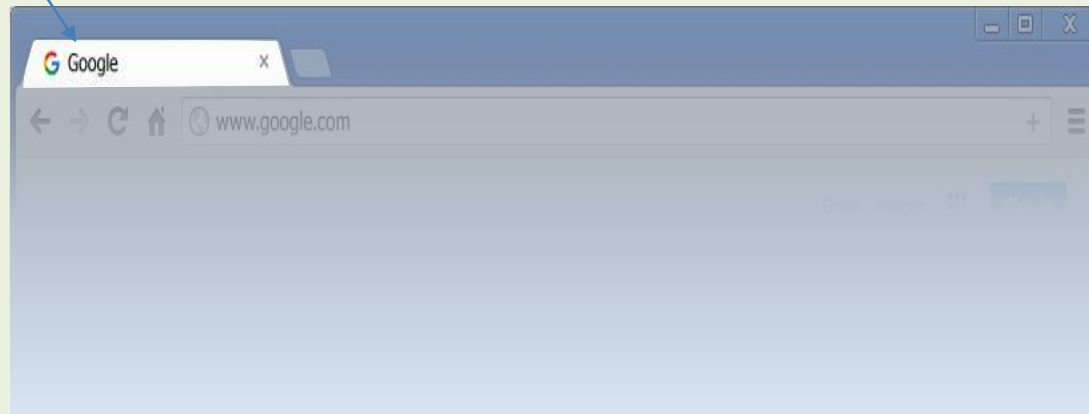
<head>

<title>

It defines a title
in the **browser
toolbar.**

It provides a title
for the page
when it is added
to **favorites.**

It displays a title
for the page in
**search-engine
results.**



The HTML **<meta>** Element

<head>

<meta>

Meta

(called as *data about data*)
holds information about
other information.

<meta>

allows to provide metadata
of HTML page.

<meta>

describes a characteristic of
a web page such as
description, keywords,
“**character encoding**” etc.

<meta>

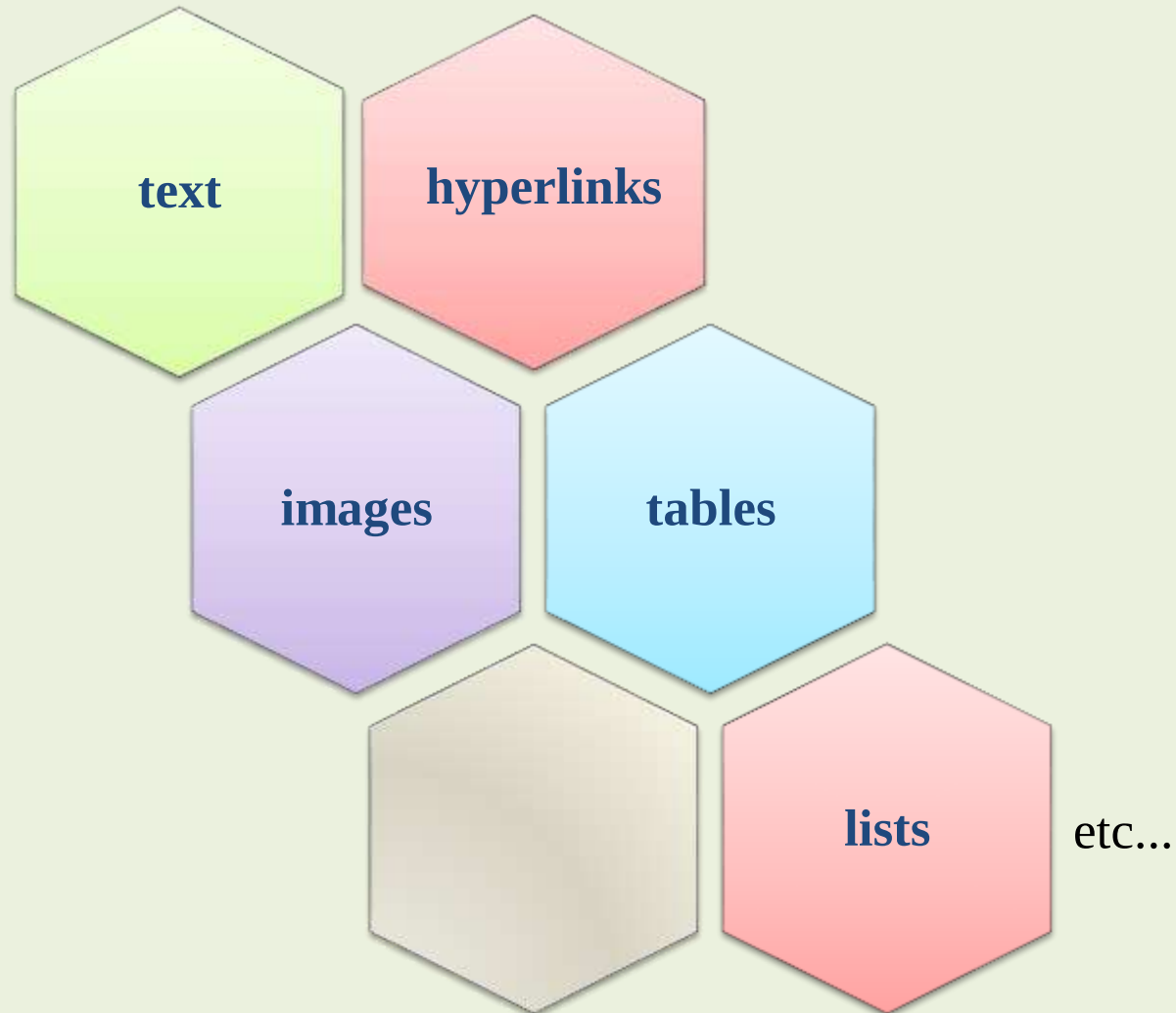
is **not used as a pair** of
opening and closing tags
and its' value *do not emerge*
on the page.

Internal representation of **letters**,
numbers and **symbols** in a file

The HTML **<body>** Element

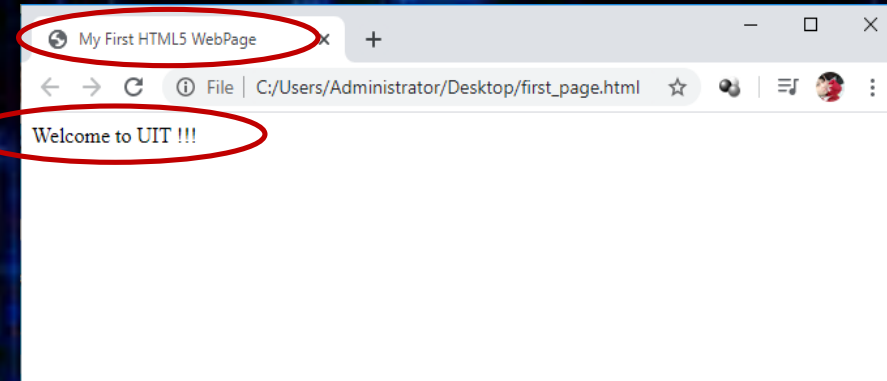
<body>

It defines the document's body and contains all contents of an HTML document, such as:



EXAMPLE

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <title>My First HTML5 WebPage</title>
    <meta charset="utf-8">
  </head>
  <body>
    Welcome to UIT !!!
  </body>
</html>
```



HTML Elements

Elements	Attribute	Description
<code><!DOCTYPE html></code>		- To declare which document type of HTML is using (HTML5, in this case). - Note: All HTML documents must start with a <code><!DOCTYPE></code>
<code><html> </html></code>	<code>lang="en"</code>	- It contains all other HTML elements (except doctype) - <code>lang</code> attribute is to inform the language of the document.
<code><head> </head></code>		- It contains all the head elements such as title, metadata, etc.
<code><title> </title></code>		- To define the title for the HTML document.
<code><meta></code>	<code>charset="utf-8"</code>	- To declare additional information about the document (Specify the character encoding with <code>charset</code> attribute)
<code><body> </body></code>		- To define the document body and contains the contents of an HTML document, such as paragraphs, images, etc.

Quiz (1)

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<title>My WebPage
```

```
<meta charset="utf-8">
```

```
</head>
```

```
<body> Welcome to UIT !!!
```



Quiz (2)

Every HTML elements always need to be closed.

(i) True

(ii) False

Some HTML elements don't need to be closed.

These are called void elements

e.g., `<meta>`





2.2

Usage of HTML Element

The HTML heading Element

- defines with the `<h1>` to `<h6>` tags
- use opening and closing tags

Syntax,

```
<h1>.....</h1>
```

```
<h2>.....</h2>
```

```
<h3>.....</h3>
```

```
<h4>.....</h4>
```

```
<h5>.....</h5>
```

```
<h6>.....</h6>
```

`<body>`

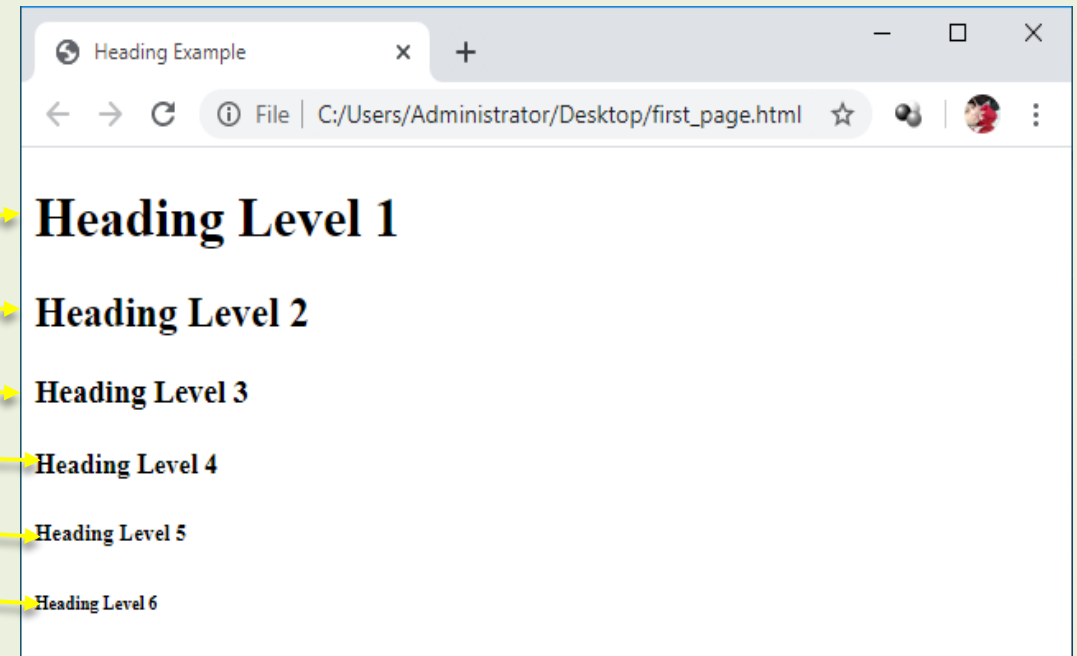
`<h1>`

Heading Element Example

<body>

<h1>

```
Untitled - Notepad
File Edit Format View Help
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Heading Example</title>
  <meta charset="utf-8" >
</head>
<body>
  <h1>Heading Level 1</h1>
  <h2>Heading Level 2</h2>
  <h3>Heading Level 3</h3>
  <h4>Heading Level 4</h4>
  <h5>Heading Level 5</h5>
  <h6>Heading Level 6</h6>
</body>
</html>
```



The HTML **paragraph** Element

<body>

<p>

- **<p> tag** defines a paragraph
- displays as a block
- will appear with empty space above and below it
- use opening and closing tags

Syntax,

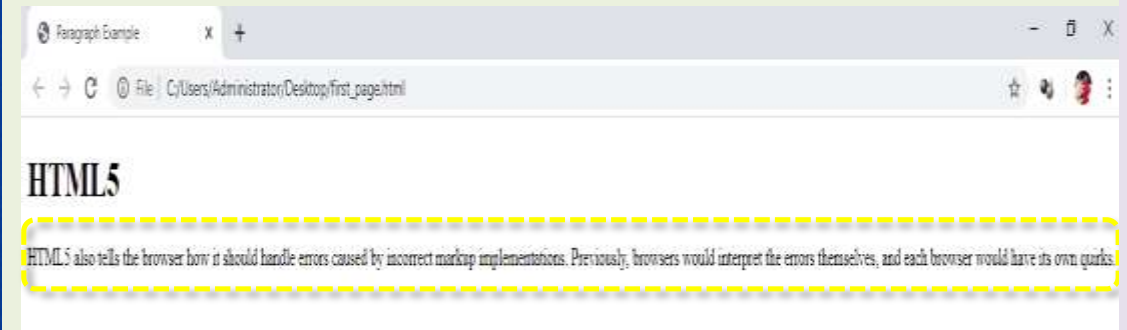
<p>.....</p>

Paragraph Element Example

<body>

<p>

```
Untitled - Notepad
File Edit Format View Help
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Paragraph Example </title>
  <meta charset="utf-8" >
</head>
<body>
  <h1>HTML5</h1>
  <p>HTML5 also tells the browser how it should
  handle errors caused by incorrect markup
  implementations. Previously, browsers would
  interpret the errors themselves, and each
  browser would have its own quirks.
</p>
</body>
</html>
```



The HTML Line **break** Element

`<body>`

`<br>`

- **
 tag** inserts a single line break.
- **
 tag** is a void element which means that it has no end tag.

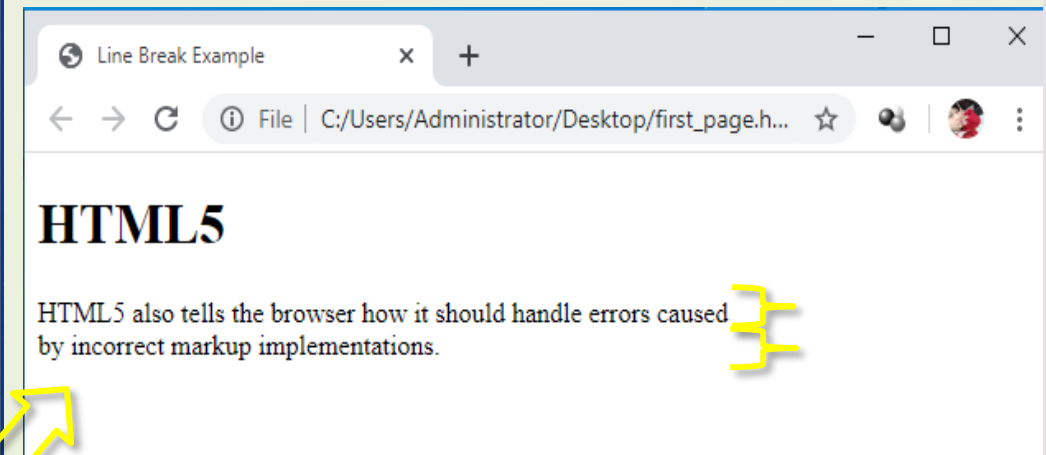
Line Break Element Example

`<body>`

`<br>`

```
Untitled - Notepad
File Edit Format View Help

<!DOCTYPE html>
<html lang="en">
<head>
  <title> Line Break Example </title>
  <meta charset="utf-8" >
</head>
<body>
<h1>HTML5</h1>
  <p>HTML5 also tells the browser how it should handle
  errors caused
  <br>
  by incorrect markup implementations.
</p>
</body>
</html>
```



The HTML **blockquote** Element

<body>

**
**

- The **<blockquote>** tag specifies a section that is quoted from another source.

Syntax,

<blockquote>.....</blockquote>

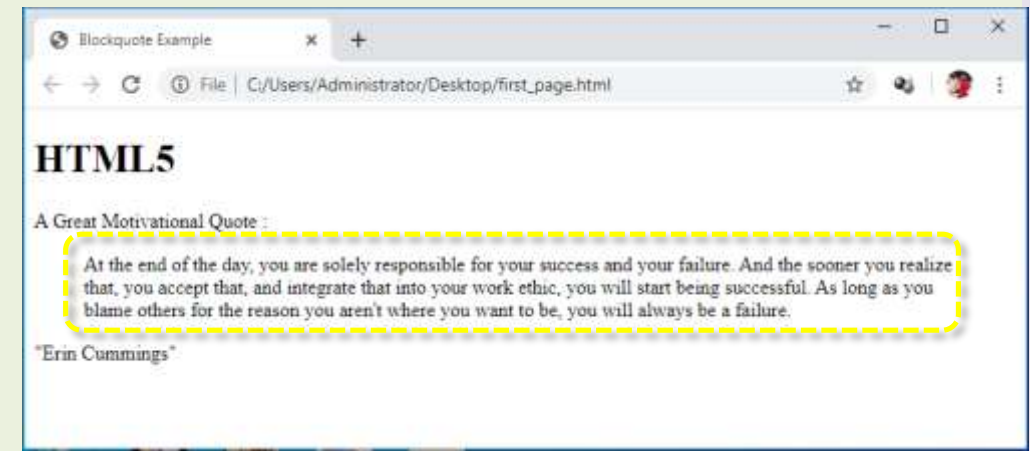
Blockquote Element Example

<body>

<blockquote>

```

Untitled - Notepad
File Edit Format View Help
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Paragraph Example </title>
  <meta charset="utf-8" >
</head>
<body>
<h1>HTML5</h1>
  <p> A Great Motivational Quote : </p>
  <blockquote> At the end of the day, you are solely
    responsible for your success and your failure. And
    the sooner you realize that, you accept that, and
    integrate that into your work ethic, you will start
    being successful. As long as you blame others for the
    reason you aren't where you want to be, you will
    always be a failure. </blockquote>
  <p> "Erin Cummings" </p>
</body>
</html>
  
```



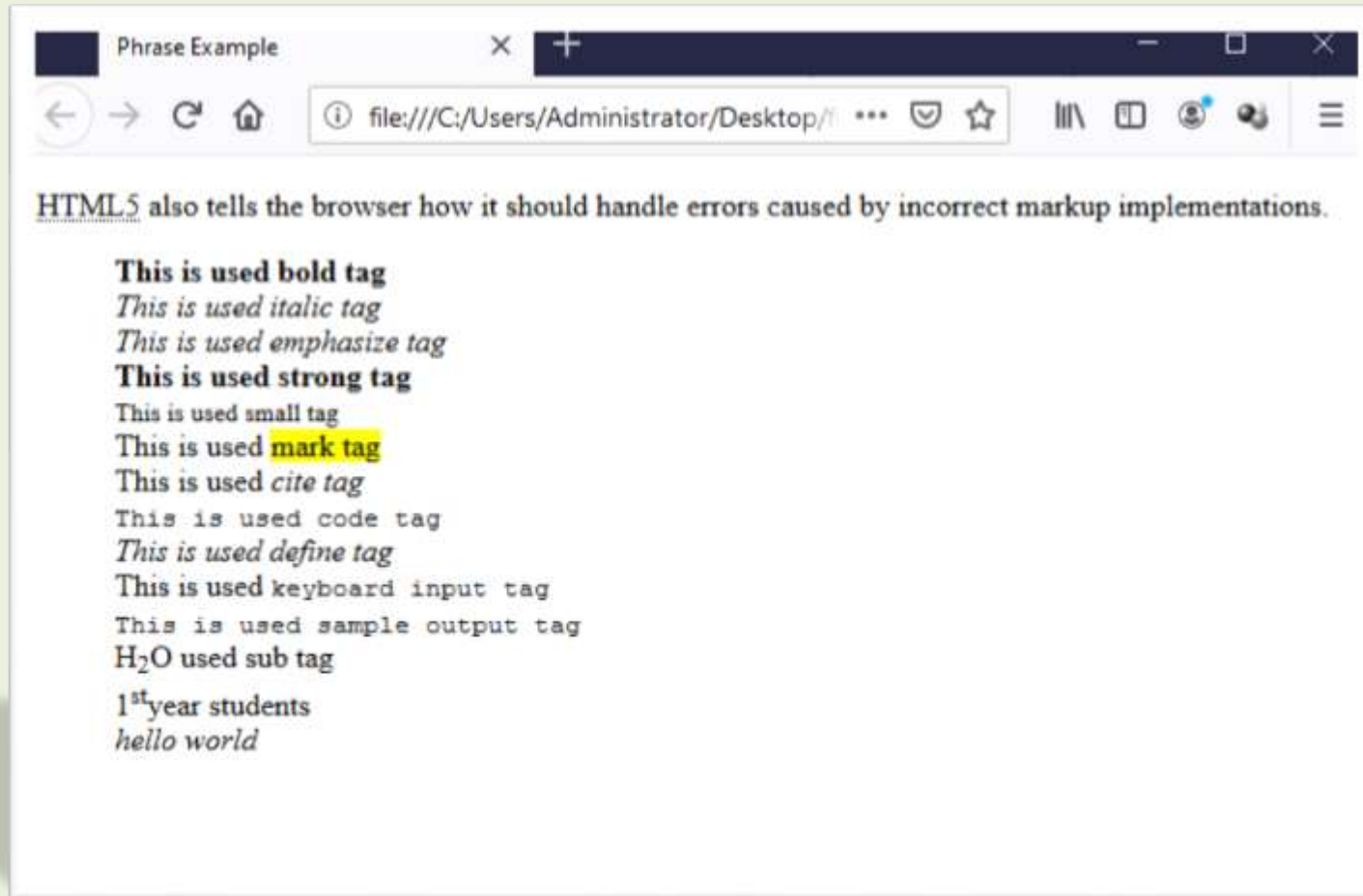
Phrase Elements	Description
<code></code> <code></code>	- To display the text in bold
<code><i></code> <code></i></code>	- To display the text in italic
<code></code> <code></code>	- To define emphasized text and the content inside is typically displayed in italic.
<code></code> <code></code>	- To define text with strong importance and the content inside is typically displayed in bold.
<code><small></code> <code></small></code>	- To define smaller text (like copyright and other side-comments).
<code><mark></code> <code></mark></code>	- To define the text that should be marked or highlighted.
<code><cite></code> <code></cite></code>	- To define the title of a creative work (e.g. a book, a poem, a song, a movie, etc).
<code><code></code> <code></code></code>	- To define a piece of computer code and the content inside is displayed in the browser's default monospace font.
<code><dfn></code> <code></dfn></code>	- “Definition element“ to specify a term that is going to be defined within the content.
<code><kbd></code> <code></kbd></code>	- To define keyboard input and the content inside is displayed in the browser's default monospace font.
<code><samp></code> <code></samp></code>	- To define sample output from a computer program and the content inside is displayed in the browser's default monospace font.
<code><sub></code> <code></sub></code>	- To define subscript text and appears half a character below the normal line
<code><sup></code> ²⁴ <code></sup></code>	- To define superscript text and appears half a character above the normal line

Phrase Elements Example

```
<!DOCTYPE html>
<html lang="en">
<head>
<title>Phrase Example </title>
<meta charset="utf-8" >
</head>
<body>
<p><abbr title="Hyper Marked up
  Language">HTML5</abbr> also tells the browser
  how it should handle errors caused by
  incorrect markup implementations.
</p>
<blockquote> <b>This is used bold tag</b><br>
  <i>This is used italic tag</i><br>
  <em>This is used emphasize tag</em><br>
  <strong>This is used strong tag</strong><br>
```

```
<small>This is used small tag</small><br>
This is used <mark>mark tag</mark><br>
This is used <cite>cite tag</cite><br>
<code>This is used code tag</code><br>
<dfn>This is used define tag</dfn><br>
This is used <kbd>keyboard input
tag</kbd><br>
<samp>This is used sample output tag </samp>
<br>
H<sub>2</sub></sub>O used sub tag<br>
1<sup>st</sup>year students <br>
<var>hello world</var>
</blockquote>
</body>
</html>
```

Phrase Elements Example



Unordered List

- ✓ ** tag** defines an unordered (bulleted) list
- ✓ Use the `<ul>` tag together with the `<li>` tag to create unordered lists.

Syntax,

```
<ul>  
    <li> ... </li>  
    <li> ... </li>  
    <li> ... </li>  
</ul>
```

`<body>`

`<ul>`

`<li>`

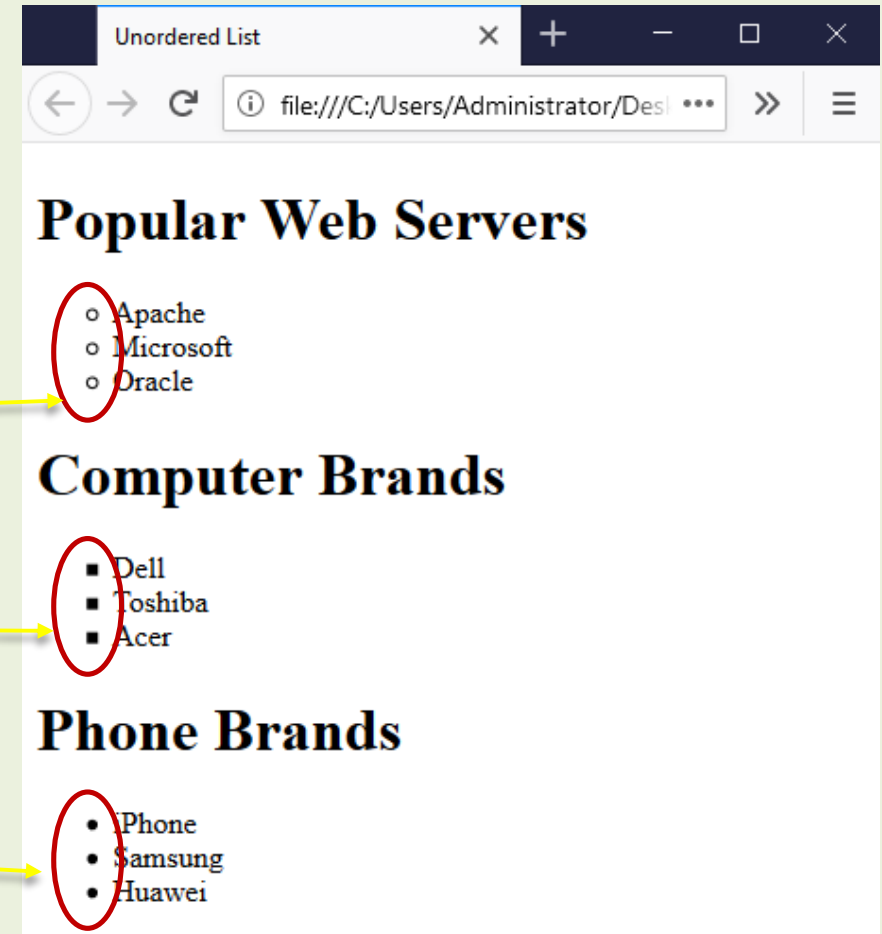
Unordered List Example

```
Untitled - Notepad
File Edit Format View Help
<!DOCTYPE html>
<html lang="en">
<head>
<title> Unordered List</title>
<meta charset="utf-8" >
</head>
<body>
<h1> Popular Web Servers</h1>
<ul type="circle">
  <li>Apache </li> <li>Microsoft</li> <li>Oracle </li>
</ul>
<h1> Computer Brands</h1>
<ul type="square">
  <li>Dell</li> <li>Toshiba</li> <li>Acer</li>
</ul>
<h1> Phone Brands</h1>
<ul>
  <li>iPhone</li> <li>Samsung</li> <li>Huawei</li>
</ul>
</body>
</html>
```

Circle

Square

Default



Ordered List

- `<ol>` tag defines **an ordered list** that can be **numerical** or **alphabetical**
- Use the `<ol>` tag together with the `<li>` tag to create ordered lists.

Syntax,

```
<ol>  
  <li> ... </li>  
  <li> ... </li>  
  <li> ... </li>  
</ol>
```

`<body>`

`<ul>`

`<li>`

Ordered List Example

```
Untitled - Notepad
File Edit Format View Help
<!DOCTYPE html>
<html lang="en">
<head>
<title> Unordered List</title>
<meta charset="utf-8" >
</head>
<body>
<h1> Popular Web Servers</h1>
<ol>
  <li>Apache </li> <li>Microsoft</li> <li>Oracle </li>
</ol>
<h1> Computer Brands</h1>
<ol type=a>
  <li>Dell</li> <li>Toshiba</li> <li>Acer</li>
</ol>
<h1> Phone Brands</h1>
<ol type=i>
  <li>iPhone</li> <li>Samsung</li> <li>Huawei</li>
</ol>
</body>
</html>
```



Description List

<dl> tag defines a description list and used in conjunction with :

<dt> (defines terms/names) and

<dd>(describes each term/name)

Note: Inside a **<dd>** tag you can put paragraphs, line breaks, images, links, lists, etc.

<body>

<dl>

<dt>

<dd>

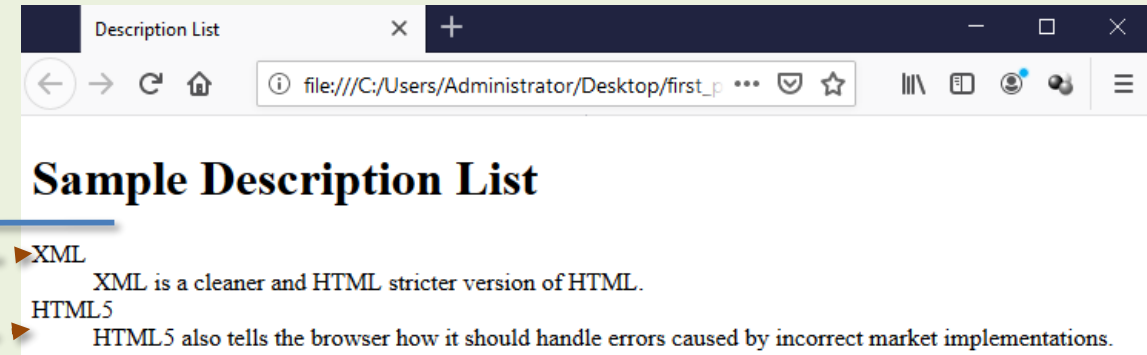
Description List Example

Untitled - Notepad

File Edit Format View Help

```
<!DOCTYPE html>
<html lang="en">
<head>
<title> Description List</title>
<meta charset="utf-8" >
</head>
<body>
<h1> Sample Description List</h1>
<dl>
  <dt> XML</dt>
  <dd> XML is a cleaner and HTML stricter version of HTML.</dd>
  <dt> HTML5</dt>
  <dd> HTML5 also tells the browser how it should handle errors
      caused by incorrect market implementations.</dd>
</dl>
</body>
</html>
```

DL



Elements	Description
<code><h1></code> <code></h1></code> <code><h2></code> <code></h2></code> <code><h3></code> <code></h3></code> <code><h4></code> <code></h4></code> <code><h5></code> <code></h5></code> <code><h6></code> <code></h6></code>	- Six different HTML headings: <code><h1></code><code><h2></code><code><h3></code><code><h4></code><code><h5></code><code><h6></code> - <code><h1></code> defines the most important heading. <code><h6></code> defines the least important heading.
<code><p></code> <code></p></code>	- To define a paragraph. A single blank line is automatically added before and after each <code><p></code> element.
<code>
</code> <code></br></code>	- To insert a single line break.
<code><blockquote></code> <code></blockquote></code>	- To specify a section that is quoted from another source.
<code></code> <code></code>	- To define (together with the <code></code> tag) to create unordered/ bulleted lists.
<code></code> <code></code>	- To define (together with the <code></code> tag) to create ordered lists in numerical or alphabetical.
<code></code> <code></code>	- To define a list item.
<code><dl></code> <code></dl></code>	- To define a description list used in conjunction with <code><dt></code> and <code><dd></code> .
<code><dt></code> <code></dt></code>	- To define a term/name in a description list.
<code><dd></code> <code></dd></code>	- To describe a term/name in a description list.

Quiz (1)

Choose the correct HTML tag for the **smallest** heading:

- (i) `<h6>`
- (ii) `<h1>`
- (iii) `<h3>`



Quiz (2)

Which of these tags are all ordered list tags?

(i) `<ol><li></li></ol>`

(ii) `<ul><li></li></ul>`

(iii) `<tr><td></td></tr>`

(iv) `<dl><dt><dd>`



Assignment (1)

- To do the html assignment, you firstly need to use a simple editor like Notepad.
- Test a simple web page with the basic HTML elements so far.
- Then, save it with an .html or .htm extension.
- After you finished, open it with your common browser.
- Your web page should have the following elements:

<h1> for a header	Other headers:	 for break
<p> for paragraph	 for bold	Italics
Unordered list	Blockquote	<sub> , <sup>
Ordered List	Description List	Some Phrase Elements





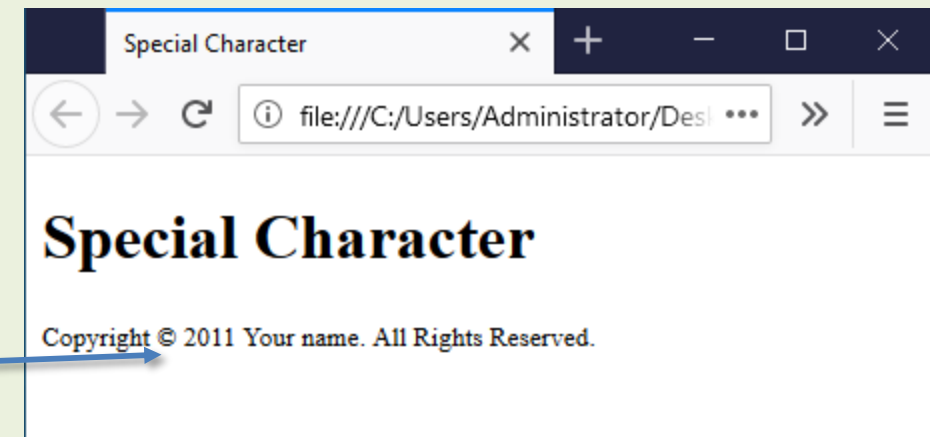
2.3

Special Characters and Grouping Element

Special Character Example

```
Untitled - Notepad
File Edit Format View Help

<!DOCTYPE html>
<html lang="en">
<head>
  <title>Special Character</title>
  <meta charset="utf-8" >
</head>
<body>
  <h1>Special Character</h1>
  <p>
    <small> Copyright &copy; 2011 Your name. All Rights
    Reserved.</small>
  </p>
</body>
</html>
```



Div Element

A **<div>** element defines a *division* or a *section* in an HTML document.

It **begins** with a **<div> tag** and **ends** with a **</div>tag**

It is used when you need to format an area of a web page, such as a **logo, navigation** or **footer area**.

<body>

<div>

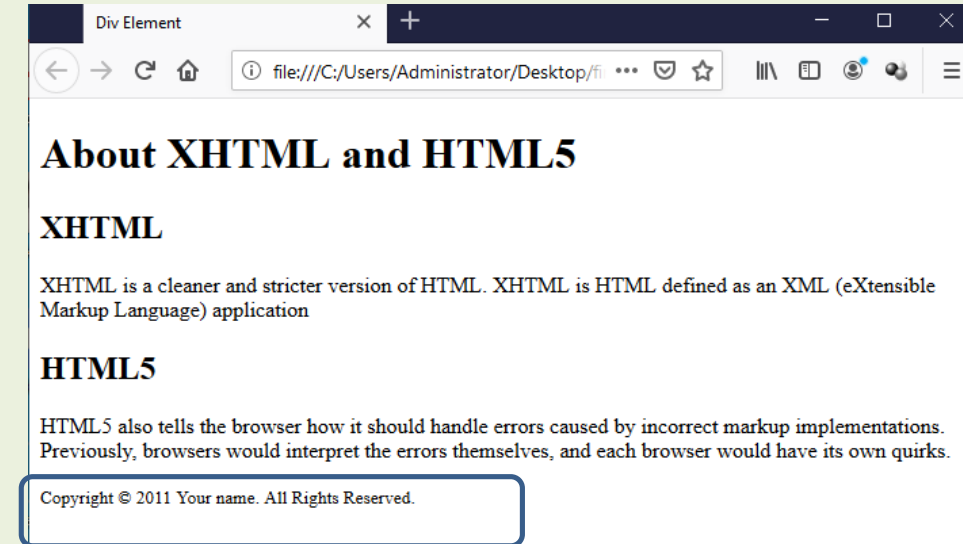
Div Element Example

File Edit Format View Help

```

<!DOCTYPE html>
<html lang="en">
<head>
  <title>Div Element</title>
  <meta charset="utf-8" >
</head>
<body>
  <h1>About XHTML and HTML5 </h1> <h2>XHTML</h2>
  <p>XHTML is a cleaner and stricter version of HTML. XHTML is
  HTML defined as an XML (eXtensible Markup Language)
  application</p>
  <h2>HTML5</h2>
  <p>HTML5 also tells the browser how it should handle errors
  caused by incorrect markup implementations. Previously,
  browsers would interpret the errors themselves, and each
  browser would have its own quirks.</p>
  <div>
    <small>Copyright &copy; 2011 Your name. All Rights Reserved.
  </small>
  </div>
</body>
</html>

```



Anchor Element

<body>

<a>

<a> defines a **hyperlink**, which is used to link from one page to another.

href is the most important attribute of **<a>**, which indicates the **link's** destination.

begins with **<a>** tag and ends with **** tag

Anchor Element Example

File Edit Format View Help

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Anchor Element</title>
  <meta charset="utf-8" >
</head>
<body>
  <h1>About Anchor Element </h1>
  <a href="first_page.html">Anchor Element Example</a>
</body>
</html>
```

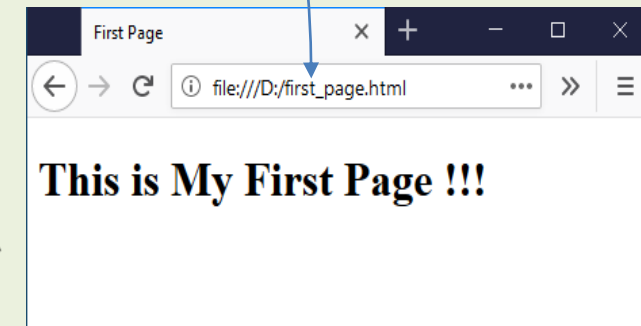
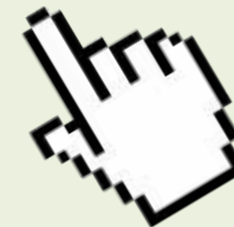


Image Element

- 
- ✓ Configures **graphics** on a web page.
 - ✓ These graphics can be **photographs, banners, company logos, navigation buttons**, and so on.
 - ✓ Use **** for HTML5 syntax.

```

```

any file name of image (.png, .gif , .jpg)

Attributes with element and align Attribute Options

Attribute	Description	Example
src	Tells the browser where to find the image	<code></code>
alt	Text description of the image	<code></code>
height	Image's height always shown in pixels	<code></code>
width	Image's width always shown in pixels	
align	Align an image within the page	<code></code>
border	A border is a line that can appear around an image or other element.	<code></code>
hspace vspace	The hspace and vspace attributes can be used to control the amount of white space around an image.	<code></code>

top	The top of the image is aligned with top of the current line of text.
middle	The middle of the image is aligned with the current text baseline.
bottom	The bottom of the image is aligned with the baseline of the current line of text (the default), which usually results in images rising above the text.
left	The image is aligned to the left side of the containing window or element and any text flows around it.
right	The image is aligned to the right side of the containing window or element and any text flows around it.

Image Link

File Edit Format View Help

```
<!DOCTYPE html>
<html>
  <head>
    ...
  </head>
  <body>
    <a href="example.html" title="Click here to return to
      the home page" >
      
    </a >
  </body>
</html>
```

Image Map

- Use the `<map>` tag to define an **image-map**.
- An image-map is an image with clickable areas.

- The name attribute of the `<map>` tag is associated with the `<img>`'s **usemap attribute** and creates a relationship between the image and map.

- The `<map>` element contains a number of **`<area>` elements**, that defines the **clickable areas** in the image-map.

Usage of Image Map

- Multiple links can be set up in one image by the use of image map function

``

... use '#' before
map name

- `<map>...</map>` used for setting link in the map.

`<map name="name of the map" >`

`<area ..... >`

`</map>`

... write the
map name

Attribute of <Area>

Attribute	value	Example
shape	rect	<code><area shape="rect"</code>
	circle	
coords	a,b,c,d	<code>coords="12, 50, 175, 240" href="anc.html" /></code>
	a,b,c	<code><area shape="circle"</code>
href	Specify the link	
		<code>coords="290, 90, 75" href="base.html"/></code>

Example image map: **map.html** with related web pages

```

Untitled - Notepad
File Edit Format View Help

<!DOCTYPE html>
<html>
  <head>...</head>
  <body>
    <h1>The map and area elements</h1>

    <p>Click on the computer, the phone, or the cup of coffee to go to
    a new page and read more about the topic:</p>

    <map name="workmap">
      <area shape="rect" coords="34,44,270,350" alt="Computer"
      href="computer.html">
      <area shape="rect" coords="290,172,333,250" alt="Phone"
      href="phone.html">
      <area shape="circle" coords="337,300,44" alt="Coffee"
      href="coffee.html">
    </map>
  </body>
</html>
  
```

computer.html

```

<html>
  <head>...</head>
  <body>
    
  </body>
</html>
  
```

phone.html

```

<html>
  <head>...</head>
  <body>
    
  </body>
</html>
  
```

coffee.html

```

<html>
  <head>...</head>
  <body>
    
  </body>
</html>
  
```

Images map and display web page according to the area clicked on the map.

workplace.jpg



Image Map



Web pages



computer.html



phone.html



coffee.html

HTML **<link>** Tag and its Attributes

- The **<link>** tag defines the relationship between the current document and an external resource.
- The **<link>** tag is often used to link to external style sheets.

```
<link rel="stylesheet"
href="styles.css">
```

attribute	Value	Description
href	URL	Specifies the destination of the linked document
rel	stylesheet, alternate, archives, author, bookmark, external, first, help, icon, last, license, next, nofollow, noreferrer, pingback, prefetch, prev, search, sidebar, tag, up	Required. Specifies the relationship between the current document and the linked document
sizes	HeightxWidth	Specifies the size of the linked resource. Only for rel="icon"
type	media_type	Specifies the media type of the linked document

Quiz (1)

Which attribute contains the `<img>` tags tag?

- (i) src
- (ii) href
- (iii) td



Quiz (2)

```
<map name="workmap">  
  < --- shape="rect" coords="34,44,270,350"  
  alt="Computer" href="computer.htm">  
</map>
```

(i) src

(ii) area

(iii) tr



Assignment (2)

- Your HTML5 page will indicate that it is in the UTF-8 character set and the title tag must contain your name.
- You must have
 - some elements like p, div, h1, ol and ul
 - three li tags in either ol or ul element or both
 - some bold and italics text in the document
 - a link to your previous assignment web page using anchor element
 - insert your favourite picture using img element and identify its width and height
 - insert one more favourite picture, use the <map> element with at least two <area> and then link to your tested html pages using href attribute



